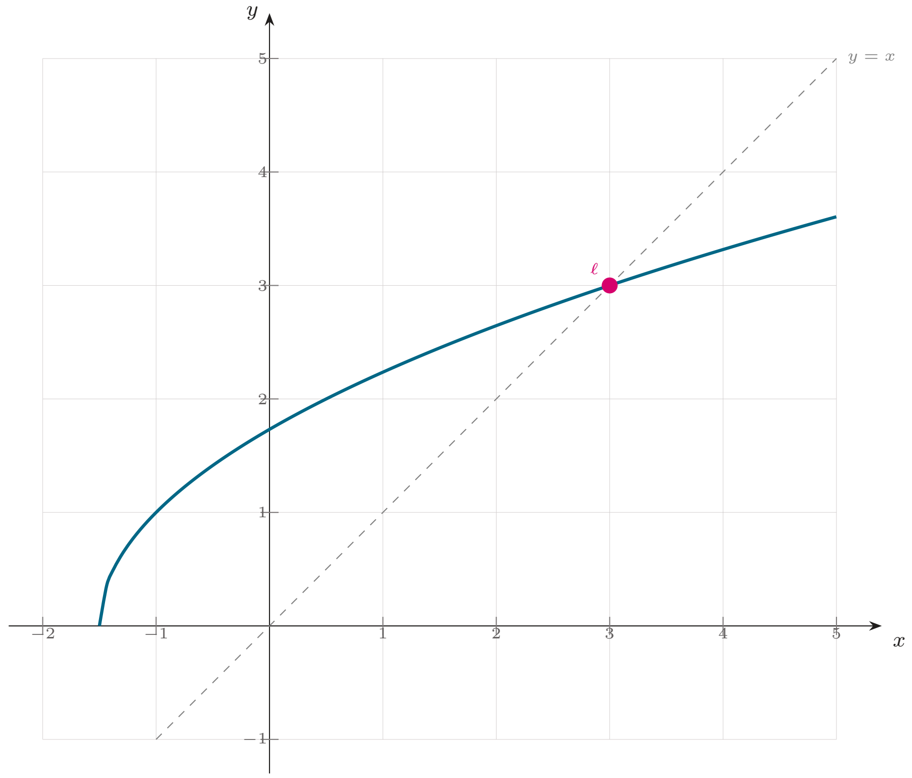
# Corrigé — Exercice 14

**A. 1)** $f'(x) = \dfrac{1}{\sqrt{2x+3}} > 0$ : croissante. **2)** $f(3) = \sqrt{9} = 3$; $f([1,3]) = [\sqrt{5}, 3] \subset [1,3]$. **B. 3)** $u_0 = 1 \in [1,3]$; si $1 \leq u_n \leq 3$ alors $u_{n+1} = f(u_n) \in [1,3]$. **4)** $u_{n+1}^2 - u_n^2 = 2u_n + 3 - u_n^2 = -(u_n - 3)(u_n + 1) \geq 0$ : croissante. **5)** croissante majorée par $3$ : converge ; $\ell = \sqrt{2\ell + 3} \Rightarrow \ell = 3$. **6)** $3 - u_{n+1} = \dfrac{2(3-u_n)}{3+u_{n+1}} \leq \dfrac{2(3-u_n)}{4} = \dfrac{1}{2}(3 - u_n)$.




$\mathcal{C}_f : y = \sqrt{2x+3}$, $y = x$ et le point fixe $\ell = 3$ (Exercice 14).